Anshuman Mondal

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Education

Indian Institute of Technology Guwahati (IITG) **2023 – 2027**
Bachelor of Technology *in Electronics and Communication Engineering* ***(ECE)*** *(CGPA :* ***8.32/10****)*

Sri Chaitanya Junior College, Hyderabad **2023**
Senior Secondary Education (Telangana State Board of Intermediate Education, TSBIE) *(Percentage :* ***98.40%****)*

The Hyderabad Public School, Ramanthapur (HPSR) **2021**
Secondary Education (Central Board of Secondary Examination, CBSE) *(Percentage :* ***98.80%****)*

Experience

Denso International India Pvt. Ltd. **May 2026 – Present**
AI R&D - Mobility Electronics Components

- Engineered a simulation pipeline using **Isaac Lab**, collecting **400+** demonstrations (**153K+** frames) to train and evaluate the **ACT control model** over **150K+** steps on **Franka Panda peg-in-hole** tasks with a **Robotiq gripper**.
- Collaborated seamlessly with teams in Japan under the **One DENSO** framework, internalizing ***Kaizen*** (continuous improvement) and ***monozukuri*** (***craftsmanship***) principles through weekly technical syncs and agile R&D execution.

Projects

The Arctic University of Norway, UiT **May 2025 – July 2025**
Research Intern - Dr. Anowarul Habib | LOR | Conference Paper *ASI Lab, Norway*

- Developed Kalman filter-based state space models for correlated time-series reconstruction in carbon composites, enabling structural health monitoring and robust damage characterization from noisy ultrasonic data.
- Presented a peer-reviewed poster on *"Reconstruction of the corrupted signal in carbon composite using Kalman Filter"* at the **Proceedings of Symposium on UltraSonic Electronics (USE)**, **2025**, **Japan**

Indian Institute of Technology, Guwahati **Sept 2024 – Mar 2025**
Research Intern - Prof. Prabir Barooah | Testimonial *India*

- Developed a robust preprocessing pipeline using **OpenCV** and **NumPy** for noisy images of **30+** tea leaf cultivars.
- Performed **foreground isolation** and extracted A4-aligned, shadow-free leaf crops from **3,000+** phone-captured field images with ≥**95%** success across **varied lighting**, camera resolutions, capturing angles, and **shadow orientations**.

Hanyang University **Nov 2025 – Dec 2025**
Research Intern - Prof. Frank Chung-Hoon Rhee *South Korea (On-Site)*

- Generated **General Type-2 fuzzy membership functions** by modeling uncertainty through repeated data deletion.
- Extended the framework to multidimensional data using **Gaussian Mixture Models**, achieving improved classification accuracy over **Type-1** and **Interval Type 2** methods with a **Feedforward Neural Network (FFNN)**.

Achievements

ML Summer School	Among top **3,000** of **0.13 Million+** candidates for **Amazon ML Summer School**.	**2026**
Machine Learning	Ranked ***1 out of 165*** participants in **8 week** Summer course for AI by IITG.ai club.	**2025**
ATF Stage-2 Candidate	Placed in **top 4K out of 20K+** applicants in *Technology Fellowship*.	**2024**
JEE Advanced	Secured an **All India Rank of 1597**, among 0.18 million candidates (top *0.88 percent).*	**2023**
JEE Main	Earned an **All India Rank of 1866**, among 1.1 million candidates (top *0.1 percent*).	**2023**
KVPY, IISc Bangalore	Attained an **All India Rank of 1487** under the **SA Stream**.	**2023**

Technical Skills

- **Programming**: C/C++, Python
- **Web Development**: HTML, CSS, React
- **ML Frameworks**: Scikit-learn, PyTorch, Tensorflow
- **Database Management**: MySQL, Firebase
- **Tools/Libraries**: Numpy, Pandas, Matplotlib
- **Miscellaneous**: Git, AWS, Linux, HuggingFace, MATLAB

Extracurricular Activities

Collaborative Video Making Program (CVMP) **2024**
Represented IIT Guwahati in an international video project with ***Gifu University (Japan)*** *&* ***UKM (Malaysia)***

Centre for Cellular and Molecular Biology (CSIR-CCMB) **2020**
Young Innovators Program (YIP) at ***Council of Scientific and Industrial Research (CSIR)*** *laboratory.Hyderabad, India*